Tutorial 2
STRESS TRANSFORMATIONS

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1. The centric force P is applied to a short post as shown in Fig. 1. Knowing that the stress on plane a-a are $\sigma = -15\,\text{MPa}$ and $\tau = 5\,\text{MPa}$, determine (a) the angle β that plane a-a forms with the horizontal,(b) the maximum compressive stress in the post.

2. For the state of plane stress shown in Fig. 2. determine (a) the value of τ_{xy}, for which the in-plane shearing stress parallel to the weld is zero, (b) the corresponding principal stresses

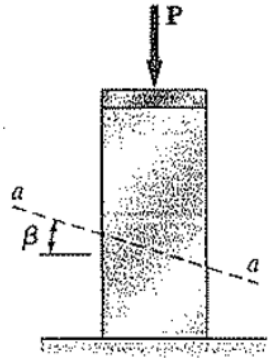


Fig. 1

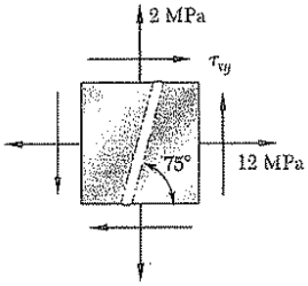


Fig. 2

3. A point on a thin plate is subjected to the two successive states of stress shown. Determine the resultant state of stress represented on the element oriented as shown in the Fig. 3.

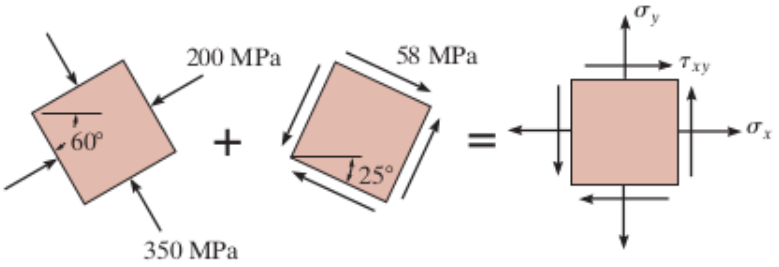


Fig. 3

4. A paper tube is formed by rolling a paper strip in a spiral and then gluing the edges together as shown in Fig. 4. Determine the shear stress acting along the seam, which is at 30° from the vertical, when the tube is subjected to an axial force of 10 N. The paper is 1 mm thick and the tube has an outer diameter of 30 mm.

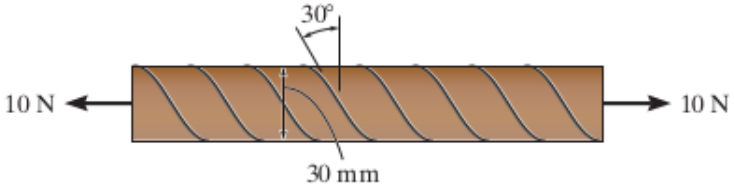


Fig. 4

5. The stress acting on two planes at a point is shown in Fig. 5. Determine the normal stress σ_b and the principal stresses at the point.

6. For the element shown in Fig 6, determine the range of values of τ_{xy} for which the maximum tensile stress is equal to or less than 60 MPa.

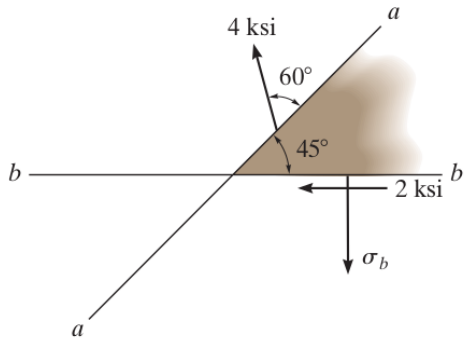


Fig. 5

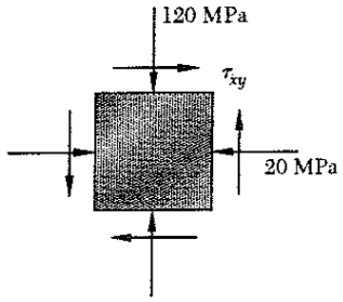


Fig. 6

7. The square steel plate has a thickness of 0.5 in. and is subjected to the edge loading as shown in Fig. 7. Determine the principal stresses developed in the steel.

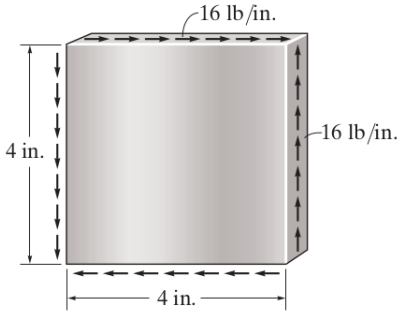


Fig. 7

8. In a plane section shown in Fig. 8 the following principal stresses are present

$$\sigma_1 = 96 \text{ MPa} \quad \text{and} \quad \sigma_2 = -52 \text{ MPa}.$$

a) Determine the stresses in sections which are inclined by $\varphi^a = 60°$ with regard to the principal axes?

b) In which section φ^b does the normal stress vanish? What are the values of the shear and normal stresses in a direction perpendicular to the direction φ^b?

c) In which directions do the maximal shear stresses appear, and what are the corresponding normal stresses?

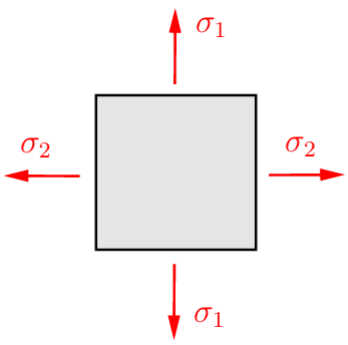


Fig. 8